

Cascaded ASR-Transformer Framework for Audio-Based Hate Speech Detection

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Abstract—The increased voice based digital communication across the social media platforms has highlighted the need for an effective detection of hate speech in spoken content. Most existing moderation systems are focused on textual data, leaving a significant gap in addressing speech based hate speech. This paper presents multistage transformer driven framework for detecting the hate speech in audio content. The proposed approach employs a Whisper based ASR model for the conversion of speech signal to text, then it is next analyzed using the fine tuned BERT model for the contextual classification. The Fine tuning is performed by updating the transformer layers and classification head to adapt semantic representations to the target task. The system has achieved the accuracy of 83.59%. A custom dataset was curated through controlled audio generation and exact preprocessing to improve the linguistic consistency. Experimental results demonstrate stable learning behavior and enhanced performance on unseen speech samples. The results have confirmed the suitability of transformer models for the detection of Hate speech in audio data.

Index Terms—Spoken Hate-Speech Detection, Audio Hate-speech Detection, Transformer Model , Offensive Language, Audio Moderation, BERT Fine tuning , Deep Learning.

I. INTRODUCTION

The digital communication between individuals has seen a rapid growth due to the unpredictable evolution of the software industry. Consequently, social media has emerged as the dominant medium for digital interaction. The use of social media is increased exponentially and made a notable surge in the users of the platforms like YouTube , Instagram , Facebook , X (formerly twitter) and etc. In which Facebook only has 3.02 billion active users worldwide and all others have minimum 2.3 to 2.8 billion active users. Due to minimal censorship and un-regulated content the use of Hate speech , Offensive language , Cyber Bullying and hate crimes are rising exponentially. According to United Nations, any communication in speech , writing or in any kind of format that uses discriminatory language for a person or a group is considered as the Hate Speech. Although Hate Speech is

always existed but this growth of digital communication and social media has increased the use impact and growth of hate speech related content. The European Union's official statistics show that 49% of individuals aged 16-29 reported seeing online messages hostile towards certain groups or individuals in 2023. It highlights that how the users are exposed to toxic content on online platforms. In early years of digital communication and social media the hate speech is almost present only in the textual content like comments and tweets. But in recent times the audio and video contents became the primary mode of digital communication, So this forced the hate speech in the audio and video content to be a major issue. As mentioned by Jinmyeong [1] , 10,501,072 channels were taken down by YouTube from July to September in 2023 in which 26,130 were regarding the Hate Speech. Therefore the manual detection and prevention of Hate Speech is not possible and the need of a fast and accurate system is stressed. So for that the Machine Learning and the Deep learning techniques will become much suitable due to their efficient algorithms and faster responses. Most of these techniques are trained on textual data and because of that there remains a research gap in detecting hate speech in audio communication. As the Hate Speech is not limited to the textual forms and occurs most of the times in spoken content and the research in that field is required. As the traditional approaches relied on the keyword lists and some machine learning techniques such as the Support Vector Machines (SVM), Logistic Regression (LG), Gradient Boost Machines (GBM), Decision Trees and K Nearest Neighbours (KNN) [2]. These techniques will struggle while detecting context, implicit abuse and variation in phrasing. In that case the Deep Learning techniques were enhanced the efficiency but they also faced limitations in understanding long-range dependencies and bidirectional context. Some of the researchers tried the End-to-End deep learning (like Wave2Vec) methods, who analyses the spoken content directly by sound wave [3]. This is good for tone but it

often misses the actual definition of the words. So we proposed a Cascaded Approach for the Hate Speech detection in audio content using ASR + BERT transformer based model. In this cascaded approach the system combines the Whisper ASR model by OPENAI, which gives the transcripts of the Audio content by converting raw waveforms into text sequences. These transcripts are then passed to BERT a Transformer dependent Deep learning technique for conceptual understating of linguistic context. We selected this specific combination because, unlike other older ASR models, Whisper is trained for 680,000 hours of Noisy data which makes it the best model for ASR in real-world scenarios. This extensive pre-training permits the model to ignore the background Noise and to focus on speech, meaning it is highly efficient in catching the exact transcription from a noisy data. Further, we fine-tuned the BERT model on transcribed text specific to our domain, and it will give the appropriate classification (Normal vs. Hate Speech). By separating the speech processing from the text analysis, this approach is better in detecting the conceptual meaning of the word and also the exact keywords, avoiding the errors common in end-to-end models. We implemented the pipeline using the Whisper Base and BERT base uncased that achieves the high accuracy and deep conceptual understanding. As there is lack of high-quality diverse audio dataset that is curated for the specific task of detecting hate speech, we faced a significant challenge in training a robust model. To solve this, we have generated the audio dataset from the textual dataset (HateXplain) [4] using text-to-speech model by google (gTTS). This allowed us to convert verified hate speech text into audio waveforms, balancing the classes effectively. We combined these synthetic samples with real-world recordings and further used it for the model training and evaluation to ensure our system could handle both clear and noisy speech inputs.

The structure of paper includes : Section II review of literature, Section III provides details of proposed methodology and model architecture. Section IV gives the results of experiments done. Section V This concludes the work by highlighting the results and the directions for future enhancements in work.

II. LITERATURE WORK

Hate speech detection has been extensively explored in text-based contexts, where early researches are based on the classical machine learning approaches like SVM, logistic regression, and Naïve Bayes [5]. The evolution toward deep learning introduced CNNs and LSTMs, which enabled richer linguistic representation learning. More recently, transformers based architectures, particularly the BERT and RoBERTa, have demonstrated competitive results due to their strong contextual encoding capabilities and bidirectional attention mechanisms [3]. These text-focused developments have affected multi modal and cascaded speech detection systems, which combine NLP-based toxicity classifiers with Automatic Speech Recognition (ASR) transcriptions [2]. Due to issues with dataset scarcity, speech pattern variability, ambient noise,

and linguistic diversity, audio-based hate speech detection is still largely unexplored, despite the fact that textual hate speech detection is well-established. Acoustic features like MFCCs, mel-spectrograms, or prosodic cues processed by CNN, GRU, and hybrid CNN-LSTM networks are used in a number of works [6] [1]. However, the limited scope and small size of current datasets frequently limit their performance. For instance, the research in [6] examines abusive speech detection for Indian languages, stressing the significance of robust features and the challenge of identifying abuse-related acoustic cues in spontaneous speech. The ADIMA corpus [7], which offers over 11,000 in-the-wild multilingual audio recordings gathered across 10 Indic languages, is a major turning point in the study of audio hate speech. Abusive, non-abusive, and code-switched utterances recorded under various circumstances are included in the dataset. The corresponding baseline models show that multilingual joint training and audio feature fusion can greatly enhance performance by utilizing CNNs, wav2vec-based embeddings, and other hybrid techniques. The authors also point out a significant drawback of cascaded ASR $\rightarrow$ NLP systems: downstream classification accuracy is directly impacted by ASR error rates, which rise disproportionately for abusive or taboo words. Recent research has concentrated on explainability and fine-grained annotation for audio hate speech in addition to dataset creation. Frame-level justifications for hate speech localization in audio are introduced by the AudioHateXplain framework, which was proposed in [1] [8]. This dataset contains both human-recorded and TTS-generated speech, allowing for a thorough assessment of the End to End (E2E) systems and the cascaded framework. According to experimental results, cascaded systems perform competitively when gold (clean) transcripts are used in place of imperfect ASR outputs, but E2E models (based on large speech encoders) outperform cascaded systems in frame-level localization. Research on multimedia hate speech detection, particularly from videos, usually employs a cascaded pipeline that extracts audio, applies ASR, and then uses machine learning or deep learning techniques to classify the resulting text. Using CNN-based feature extraction, the work in [2] shows that audio modality makes a significant contribution when paired with text and visual cues in multimodal settings. Further multimodal studies, like [6], look at how audio, ASR transcripts, and contextual metadata can be used to detect harmful situations. These studies confirm that ASR performance is crucial for cascaded models and that combining textual and acoustic data can improve robustness, especially in noisy settings. Finally, research in [1] presents an end-to-end audio-based detection pipeline using large-scale speech encoders on Korean datasets. This work reinforces the finding that direct audio modelling can outperform ASR-dependent systems when sufficient labelled audio data is available. However, such resources remain scarce for most languages, making cascaded ASR-NLP pipelines—like Whisper + BERT—practical and effective for many real-world scenarios. Overall, the literature emphasizes three dominant trends: (1) Text-based models are mature and highly accurate, but

limited when applied to noisy ASR transcripts. (2) Cascaded ASR $\rightarrow$ NLP pipelines offer strong interpretability but suffer when ASR misrecognizes toxic words. (3) End-to-end audio models excel when trained on large annotated datasets but remain resource-intensive.

III. PROPOSED METHOD

A. Dataset :

As there is lack of high-quality diverse audio dataset curated for the specific task of detecting hate speech , so we have generated the spoken-audio dataset for model training and evaluation. Our work constructs a new spoken audio dataset for the hate speech by converting well established textual hate speech resource into high quality MP3 audio samples. HateXplain [4] , a bench mark dataset for the hate speech in textual format is used to generate labeled audio dataset. The dataset creation is one of the contribution of the paper, ensuring that the final model can robustly interpret Hate speech and normal language expressed through speech.

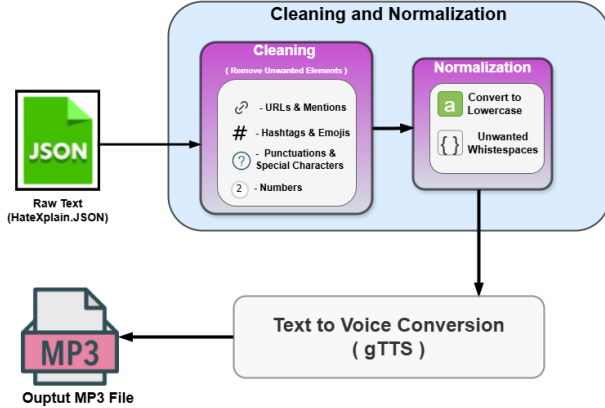


Fig. 1: Dataset Generation Pipeline

1) Dataset Preprocessing(Text Extraction and Cleaning):

The creation of Dataset begins with extracting the labelled text from the HateXplain [4] corpus. HateXplain is a benchmark dataset widely used to study hate speech and toxic language [4]. It is provided in structured JSON format , where each record contains a sequence of tokenized words (post_tokens), a set of annotator labels (annotators) and additional metadata. In this source provided the data in fragmented token lists (e.g., ["muzzie", "pigs"]). Therefore to recover the original sentence, the tokens of each entry are combined into single coherent text. The dataset includes multiple annotator opinions per comment, a majority-vote mechanism is applied to determine the final label between the three categories provided as : Normal , Hate Speech , Offensive language.

These sentences often contains some kind of noise as abbreviations , emojis , irregular spellings and multilingual fragments. This noise will affect the audio quality of the synthesized speech but also reduces the accuracy of ASR transcription as reported in prior audio-based hate speech

studies [7]. Therefore , each sentence undergoes a cleaning and normalization pipeline shown in the fig1

2) *Audio Dataset creation and validation* : After pre-processing , each cleaned sentence is converted to speech [1] using Google Text-to-Speech (gTTS) and saved in MP3 format. Due to the execution limit , the dataset was processed in nine separate batches , with each account generating approximately 1000 audio files. To maintain consistency across batches every audio file follow a systematic naming scheme as

$$Ax_xxxx_label.mp3 \quad (1)$$

In addition, a metadata CSV file is automatically generated for each batch , containing filename , assigned label , cleaned text used for synthesis. Combining all batches result in a large and diverse dataset of spoken normal , Offensive and hate speech. To guarantee that only usable audio samples proceed to training, all generated MP3 files are passed through automatic detection of Corrupted audio files. Here, Each file is programmatically loaded and analysed. Files exhibiting any issues like corrupted or unreadable MP3 frames , Zero-duration audio , completely silent output are removed from the dataset. After this dataset filtering and validation step the final dataset containing 8000 samples is prepared. In which the hate speech and offensive language classes are combine and given with class-1 and normal speech is given with the class-0, and the distribution is as follows :

TABLE I: Class-wise Distribution of the Dataset

Class	Category	No. of Samples
0	Normal	2332
1	Hate Speech	5768

B. Model Architecture and training

Our methodology integrates two major components as automatic speech recognition (ASR) and fine-tuned BERT based classifier to create a pipeline for hate/offensive speech detection from audio data. The complete system consists of following stages,

- 1) Speech-to-text conversion (Whisper model)
- 2) Text Tokenization and Embedding Construction
- 3) Transformer Encoding (12 layer BERT model)
- 4) Classification Layer

1) **Speech-to-text conversion:** The spoken utterances are inherently unstructured and not suitable for the direct use in NLP or Transformer based hate speech classifiers. After fine tuning BERT model is highly efficient in detecting the hate speech including conceptual understanding of words. As BERT only support textual data [9], there is the need for Speech-to-text conversion. Therefore we are using OPENAI's Whisper Base ASR model [10] [11] , which converts the audio signals into textual transcripts. Whisper model is highly effective because it is trained using a large collection of multilingual data, which enables it to handle Different accents , Background noises , Conversational utterances , Informal speech.

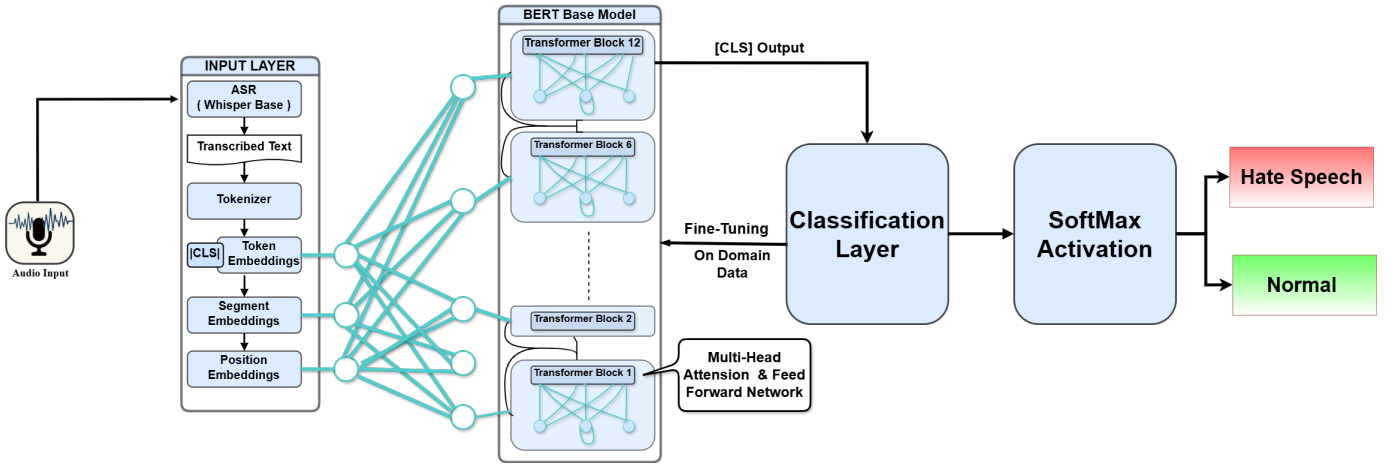


Fig. 2: Overview of the proposed cascaded architecture for audio-based hate speech detection.

2) **Tokenization and Embedding:** The textual transcript is first passed through the BERT tokenizer, which uses a WordPiece subword tokenization strategy. Instead of treating each word as a distinct unit, it breaks the text into smaller and meaningful subword segments. This allows the BERT model to efficiently handle rare words, slang, hateful variations. This subword-level decomposition helps the model generalize better across variations in speech patterns and informal language. The tokenizer performs text normalization, Lowercasing, WordPiece subword segmentation, Conversion to numerical IDs.

After tokenization, each subword token is mapped to a 768-dimensional dense vector, forms the input to the BERT Transformer layers. BERT model constructs these embeddings by summing three components:

- 1) **Token Embeddings** (T_i)
- 2) **Positional Embeddings** (P_i)
- 3) **Segment Embeddings** (S_i)

Token Embeddings capture the semantic meaning of each token or subword. This helps BERT to distinguish the patterns based on meanings. The positional Embeddings tells that where the word appears and these are there to ensure the model's understanding on grammatical structure of the sentence. The Segment Embeddings identifies that to which sentence a token belongs to. In our model we perform single sentence classification, therefore the segment IDs remain uniform. Therefore the final input embedding E_i (768 dimensional vector) at position i is computed as:

$$E_i = T_i + P_i + S_i \quad (2)$$

3) **Transformer Encoding:** The input embeddings are passed into a stack of 12 Transformer Encoding layers, where each layer is composed of the following components:

- 1) **Multi Head Self Attention (MHSA):** The mechanism enables contextual learning through attention weights, which allows each and every token to attend the all other

tokens in the sequence. The self attention operation is represented mathematically as:

$$\text{Attention}(K, Q, V) = \text{softmax}\left(\frac{K^T Q}{\sqrt{d_k}}\right) V \quad (3)$$

Where K denotes key, Q denotes query and V denotes value matrices. The d_k is the dimensionality of the key vectors.

- 2) **Feed Forward Network (FFN):** Each Transformer layer includes a two-layer fully connected feed-forward network that introduces non-linear transformations and these enables the BERT model to capture the complex semantic relationships.
- 3) **Residual Connections and Layer Normalization:** The Residual connections are there around the both attention and feed forward sublayers, followed by layer normalization, to ensure stable training and smooth gradient flow across deep Transformer layers.

4) **Classification Layer using CLS Token:** As the contextualized embeddings are passed through the multi-layer BERT encoder, the learned representations are transformed into a definitive prediction indicating whether the spoken content contains hate speech or not. For this classification task, the [CLS] token acts as a global sentence-level representation, denoted as:

$$h_{\text{CLS}} = H_0^{(12)} \quad (4)$$

where $H_0^{(12)}$ representing the hidden state corresponding to [CLS] token which is obtained from the final (12th) Transformer encoder layer. This results in a feature vector of 768-dimensions that encapsulates the semantic context of the entire input sequence.

The extracted representations are then fed through a fully connected dense classification layers followed by a activation function (softmax), defined as:

$$\hat{y} = \text{softmax}(b + Wh_{\text{CLS}}) \quad (5)$$

where W denotes the trainable weight matrix and b denotes the bias vector. This operation yields the predicted probability distribution over the two target classes: *Normal* and *Hate Speech*. The classification process aligns with the 768-dimensional to 2-dimensional mapping illustrated in Fig.2.

C. Model Training

In our work we have done fine tuning of the BERT base uncased model to predict the appropriate output for input audio provided. The fine tuning means that we have updated all the 12 Transformer layers , all multi-head attention weights , all feed-forward networks , all layer normalization parameters , CLS representation and classification head during model training. The dataset is divided into training dataset and validation or testing dataset. The training is conducted for multiple epochs on the files present in the dataset using the mini-batches. Where each batch consists around 200-260 files and the training is conducted using these batches. Batch wise sampling is done at the beginning and flow starts from whisper model , then tokenizer , then BERT Encoder and then Classifier. Cross-entropy loss is calculated for each batch. We have used the AdamW optimizer, which updates all transformer layers. After every epoch, the model is evaluated on a separate validation subset. During the training we record validation accuracy , loss curves, confusion matrix F1-score , recall and precision. The final model with the higher validation accuracy is saved.

IV. RESULTS

To validate the efficacy of the proposed cascaded architecture, we conducted a rigorous series of experiments using the synthetic audio dataset described in the Methodology section. The primary objective was to determine if decoupling the acoustic modeling (via Whisper) from the semantic classification (via BERT) could yield a robust hate speech detection system that is both accurate and interpretable. The model was trained over 5 epochs using the AdamW optimizer with a learning rate of 2×10^{-5} and a batch size of 8. The following subsections detail the quantitative performance, training stability, and qualitative explainability of the system.

A. Quantitative Performance Evaluation

As shown in the classification report[TableII], standard evaluation metrics such as precision, recall, F1-score, and accuracy are used to summarize the overall classification performance of the suggested model. The system’s overall accuracy of about 84% shows that the cascaded ASR-Transformer architecture works well for identifying binary hate speech in audio data. The weighted F1-score of 0.84 further demonstrates that the model performs consistently even when there is class imbalance, which is a typical feature of datasets containing hate speech. There is a noticeable difference in performance between the two categories when class-wise results are examined more closely. Strong confidence and consistency in identifying

the Normal class are demonstrated by the model’s precision of 0.93, recall of 0.86, and F1-score of 0.89, indicating strong confidence and consistency in identifying non-harmful speech.

The model records a recall of 0.75 for the Hate/Offensive class, indicating that a significant percentage of harmful content is accurately identified. The precision of 0.56, however, suggests that some common statements are mistakenly categorized as offensive or hateful. Crucially, giving recall for harmful content top priority is consistent with real-world moderation requirements, where missing abusive speech can have more serious repercussions than sporadic false alarms.

TABLE II: Classification Report

Class	Precision	Recall	F1-score	Support
Normal	0.93	0.86	0.89	398
HateSpeech	0.56	0.75	0.64	96
Macro Avg	0.75	0.80	0.77	494
Weighted Avg	0.86	0.84	0.84	494

B. Validation Accuracy Across Epochs

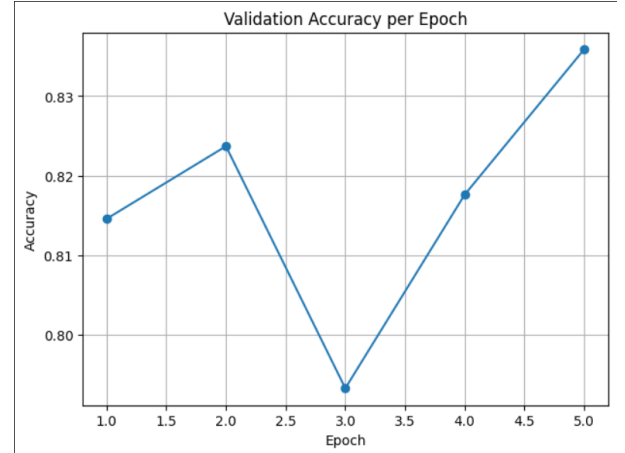


Fig. 3: Validation accuracy per epoch

The validation accuracy trend over training epochs can be used to infer the model’s learning dynamics and convergence behavior. The validation accuracy curve (fig3) illustrates the model’s generally increasing trend, with its highest accuracy of 83.59% in the last epoch. The adaptive nature of fine-tuning, where the model shifts from generic linguistic representations to task-specific decision boundaries, is responsible for the brief decline in validation accuracy around the third epoch. Transformer fine-tuning literature has extensively documented such behavior, especially when training on noisy or weakly supervised data derived from ASR outputs. The model successfully stabilizes and adjusts to the hate speech classification task, as evidenced by the subsequent recovery and increase in accuracy.

Overall, the validation accuracy curve confirms that the proposed training strategy does not suffer from early stagnation and benefits from continued optimization over multiple epochs.

C. Training and Validation Loss Analysis

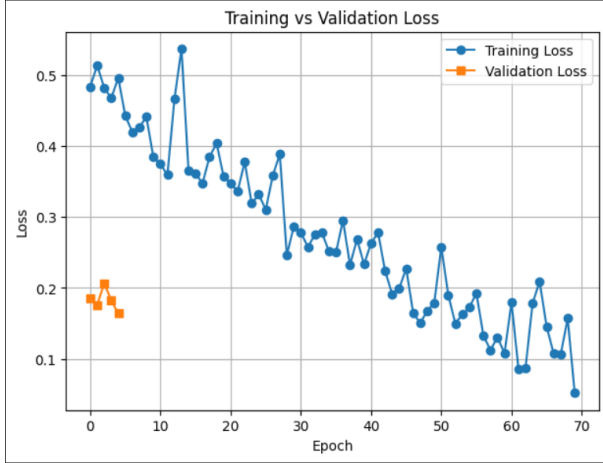


Fig. 4: Validation accuracy per epoch

The suggested Whisper-BERT cascaded model’s training behavior shows steady and effective convergence. With a validation accuracy of 82.37%, the model quickly lowers training loss to the range of 0.35–0.40 during the first epochs (Epochs 1–2). This quick convergence shows how well pre-trained BERT representations on Whisper-generated transcripts work, enabling the model to pick up explicit hate speech patterns fast. While training loss continues to decline, Epoch 3 shows a brief decline in validation accuracy (79.33%). This behavior implies that the model comes across more complicated and context-dependent samples, where it becomes difficult to discern subtle offensive intent. These brief variations frequently occur during transformer fine-tuning and usually signify optimization through intricate areas of the loss landscape as opposed to overfitting.

The model recovers in subsequent epochs, reaching its best performance at Epoch 5 with a validation accuracy of 83.59% and training loss reduced to approximately 0.05–0.10. The close alignment between decreasing training loss and improving validation accuracy confirms effective learning without memorization. Furthermore, the consistently lower and stable validation loss compared to training loss indicates strong generalization, aided by pre-trained weights and the regularization effects of transformer architectures. Overall, these results validate the robustness of the cascaded ASR-NLP pipeline for audio-based hate speech detection.

D. Confusion Matrix Analysis

A more detailed picture of prediction behavior in both classes is given by the confusion matrix. The high true negative count indicates that most Norwegian samples are correctly classified. In a similar vein, a significant number of Hate/Offensive samples are accurately identified, confirming the model’s capacity to identify detrimental patterns in spoken language. But the matrix also shows patterns of asymmetric misclassifications. The lower precision seen for the hate class

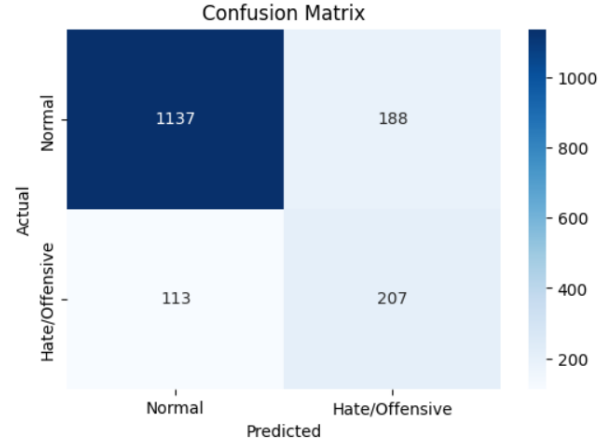


Fig. 5: Confusion Matrix

is consistent with a significant percentage of normal samples being incorrectly classified as hate/offensive. Contextual overlap, sarcasm, an aggressive tone without overt abuse, and transcription artifacts introduced by the ASR stage are likely to have an impact on this result. These challenges have been well documented in earlier audio-based hate speech research.

Despite these difficulties, the confusion matrix verifies that the model shows a significant distinction between the two classes and that misclassifications are more often the result of linguistically ambiguous situations than of random errors.

E. Per-Class Performance Comparison

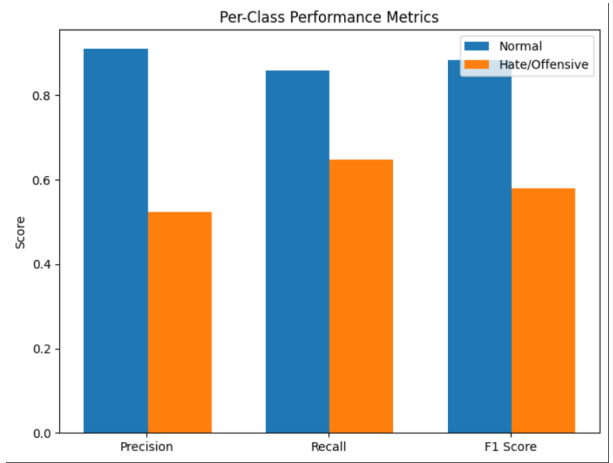


Fig. 6: Per Class Performance

The distinction between normal and hate/offensive speech detection becomes even clearer when performance metrics are visualized by class. Precision, recall, and F1-score consistently remain high for the typical class. Although the scores for the hate/offensive class are lower, they are still noteworthy. Because hate speech can be implicit, coded, or context-dependent, this imbalance reflects the inherent difficulty of hate speech detection. This difficulty is made worse by spoken

hate speech because of prosodic variation, accent differences, and noise produced by ASR. However, obtaining an F1-score above 0.6 for the hate/offensive class shows that the suggested system does not only rely on surface-level keywords to capture a substantial amount of abusive content.

Given that the lack of extensive, manually annotated audio hate speech datasets, these results are competitive with current cascaded systems documented in the literature.

F. ROC Curve and Discriminative Ability

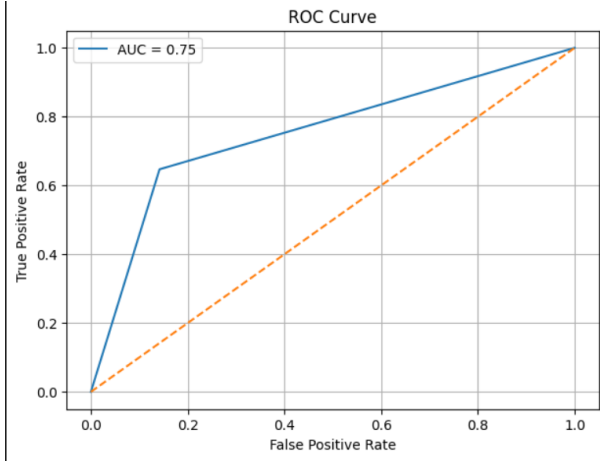


Fig. 7: ROC Curve

An overall indicator of the model's capacity for discrimination across a range of decision thresholds is provided by the receiver operating characteristic (ROC) curve. The classifier performs significantly better than random guessing and maintains a reasonable balance between true positive and false positive rates, as indicated by the obtained AUC score of 0.75. For real-world hate speech detection tasks, an AUC in this range is regarded as strong, especially when using ASR-generated transcripts instead of clean text. The ROC analysis confirms that the model's predictions are not unduly sensitive to a single threshold and that its confidence estimates are significant at different operating points. This feature is especially important in deployment scenarios where platform-specific moderation policies may require decision thresholds to be adjusted. This feature is particularly crucial for deployment scenarios because platform-specific moderation policies may require adjusting decision thresholds.

G. Discussion

Overall, the experimental findings confirm that the suggested Whisper + BERT cascaded architecture for audio-based hate speech detection is effective. Strong generalisation, consistent learning behaviour, and significant differentiation between harmful and normal speech are all displayed by the model. The observed recall-oriented behaviour is in line with the objectives of content moderation systems, even though there are still difficulties in increasing precision for the hate/offensive class. The findings also demonstrate the

usefulness of distinguishing between semantic understanding and speech recognition. The system avoids many of the drawbacks of purely end-to-end audio-based approaches by utilising BERT's deep contextual modelling capabilities and Whisper's resilience to noisy audio. The suggested framework is appropriate for practical implementation because of its modular design, which also permits interpretability, extensibility, and domain adaptation.

V. CONCLUSION AND FUTURE WORKS

This study suggested a cascaded audio-based framework for detecting hate and offensive speech that combines a refined transformer-based language model with automatic speech recognition. The system successfully captures both acoustic robustness and semantic understanding by using a BERT-based classifier for contextual analysis and the Whisper ASR model to translate speech into text. Transformer-based textual representations are well suited for handling ASR-generated transcripts in real-world audio scenarios, according to the experimental results, which show stable convergence and competitive performance.

There are still difficulties in identifying subtle and implicit hate speech, especially because of class imbalance and transcription errors, even though the suggested method achieves encouraging accuracy and generalisation. Future research will concentrate on incorporating more extensive and varied real-world audio datasets, investigating hybrid models that simultaneously learn from textual and acoustic features, and expanding the framework to detect hate speech in multiple languages and classes. Furthermore, incorporating explainability strategies can enhance openness and confidence even more, making their practical implementation in online content moderation systems possible.

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